

NLP Practice Question Bank

Complete Edition — All Questions in One File

Part A: 150 original practice MCQ & True/False questions (NLP Overview, Text Pre-Processing, Text Representations).

Part B: Verbatim questions from your uploaded Mock Midterm (2 attempts) and Quizzes 1–4.

Part C: 74 practice questions covering Lecture 5 — Sentence Embeddings, Recurrent Neural Networks (RNN), LSTM, and Word2vec (CBOW, Skip-gram, Negative Sampling, Training Process, Gensim).

Part D: Verbatim questions from your uploaded Quiz 5 and Quiz 6 attempt reviews, with full step-by-step explanations in the answer key.

#	Section	Questions
1	Lecture 1: NLP Overview	30
2	Text Pre-Processing: Introduction & Regex	23
3	Text Pre-Processing: Tokenization	8
4	Text Pre-Processing: Stop Words	5
5	Text Pre-Processing: Stemming & Lemmatization	13
6	Text Pre-Processing: POS Tagging	21
7	Text Pre-Processing: Corpus & Automatic Tagging	7
8	Text Pre-Processing: NER	4
9	Text Representations	39
	PART A TOTAL	150
10	Mid-Term Mock Exam (Attempt 1)	16
11	Mid-Term Mock Exam (Attempt 2)	16
12	Quiz 1	4
13	Quiz 2	6
14	Quiz 3	3
15	Quiz 4	2
	PART B TOTAL	47
16	Lecture 5: Sentence Embeddings (Intro)	9
17	Lecture 5: Recurrent Neural Network (RNN) & LSTM	27
18	Lecture 5: Word2vec (CBOW, Skip-gram, Neg. Sampling, Training, Gensim)	38
	PART C TOTAL	74
19	Quiz 5	2
20	Quiz 6	3
	PART D TOTAL	5

	GRAND TOTAL	276
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PART A — 150 Original Practice Questions

Lecture 1: NLP Overview

1. NLP is a subfield of:
 - A. Linguistics, Computer Science, and Artificial Intelligence
 - B. Computer science only
 - C. Artificial Intelligence only
 - D. Linguistics only
2. ELIZA was created between 1964 and 1966.
True / False
3. ELIZA was designed to imitate the responses of:
 - A. A doctor
 - B. A teacher
 - C. A lawyer
 - D. A Rogerian psychotherapist
4. ELIZA was one of the first programs capable of attempting the Turing test.
True / False
5. In the Turing Test, who plays the role of the Interrogator?
 - A. The human respondent
 - B. None of these choices
 - C. The human questioner
 - D. The computer respondent
6. In the Turing Test, the computer respondent is also called the:
 - A. Expert
 - B. AI
 - C. Interrogator
 - D. Judge
7. In the Turing Test, the human respondent is also referred to as the:
 - A. Expert
 - B. AI
 - C. Examiner
 - D. Interrogator
8. The Turing test measures the performance of an intelligent machine against that of a human being.
True / False
9. ChatGPT has been listed as a co-author in a peer-reviewed paper, according to the lecture.
True / False
10. Which stage of language processing deals with understanding the patterns present in sound and speech?
 - A. Lexical
 - B. Pragmatic

- C. Phonetical and Phonological
- D. Syntactic

11. Which stage of language processing involves understanding the structure of words and their morphemes?

- A. Discourse
- B. Semantic
- C. Morphological
- D. Phonological

12. Which stage of language processing involves understanding the part of speech of words?

- A. Syntactic
- B. Discourse
- C. Lexical
- D. Pragmatic

13. Which stage of language processing deals with the structure of sentences (grammar)?

- A. Syntactic
- B. Semantic
- C. Lexical
- D. Phonological

14. Which stage of language processing deals with the literal meaning of words and sentences?

- A. Pragmatic
- B. Morphological
- C. Semantic
- D. Discourse

15. Which stage of language processing deals with units larger than a single sentence?

- A. Syntactic
- B. Lexical
- C. Morphological
- D. Discourse

16. Which stage of language processing requires understanding the world context beyond the sentence itself?

- A. Syntactic
- B. Lexical
- C. Phonological
- D. Pragmatic

17. The sentence 'I made her duck' is a classic NLP example used to illustrate:

- A. Stop word removal
- B. Stemming
- C. Tokenization
- D. Ambiguity

18. The words 'write' and 'right' sounding the same illustrate which type of ambiguity?

- A. Semantic
- B. Pragmatic

- C. Syntactic
- D. Phonological

- 19.** The word 'bank' meaning either a financial institution or a riverbank illustrates which type of ambiguity?
- A. Word senses ambiguity
 - B. Phonological ambiguity
 - C. Part of speech ambiguity
 - D. Syntactic ambiguity
- 20.** The word 'play' being either a noun or a verb illustrates which type of ambiguity?
- A. Discourse ambiguity
 - B. Part of speech ambiguity
 - C. Word senses ambiguity
 - D. Phonological ambiguity
- 21.** The sentence 'I can see a man with a telescope' illustrates which type of ambiguity?
- A. Phonological ambiguity
 - B. Morphological ambiguity
 - C. Word senses ambiguity
 - D. Syntactic structure ambiguity
- 22.** Ambiguity is one of the main reasons that makes NLP a difficult task.
True / False
- 23.** Identifying the brand sentiment expressed across social channels is the task of:
- A. Email filtering
 - B. Chatbot
 - C. Social media monitoring
 - D. Predictive text
- 24.** Detecting spam, viruses, and malware before they reach a user is mainly the task of:
- A. Sentiment analysis
 - B. Paraphrasing
 - C. Predictive text
 - D. Email filtering
- 25.** Predictive text is often used to detect spam, viruses, and malware before they reach a user.
True / False
- 26.** Sentiment analysis is the process of reducing text size and creating a summary of the text data.
True / False
- 27.** Reducing text size and creating a summary of the text data describes:
- A. Automatic summarization
 - B. Sentiment analysis
 - C. Predictive text
 - D. Social media monitoring
- 28.** A computer program that simulates human conversation with an end user is called a:
- A. Chatbot
 - B. Parser

- C. Tokenizer
- D. Tagger

29. Suggesting words a user may wish to type next in a text field is the function of:

- A. Predictive text
- B. Stemming
- C. NER
- D. Paraphrasing

30. Helping students and professionals rewrite, edit, and change the tone of their text to improve clarity describes:

- A. Tokenization
- B. NER
- C. Sentiment analysis
- D. Paraphrasing

Text Pre-Processing: Introduction & Regex

31. Text preprocessing refers to:

- A. Compressing text files for storage
- B. Translating text from one language to another
- C. Converting human language text into machine-interpretable text
- D. Converting machine code into human language

32. Text preprocessing involves the cleaning of text before the application of NLP techniques.

True / False

33. There is one fixed order of text preprocessing techniques that must always be used regardless of the use case.

True / False

34. A regular expression is best defined as:

- A. A vector representation of a word
- B. A sequence of characters that specifies a search pattern in text
- C. A type of neural network
- D. A database query language

35. Which built-in Python package is used to work with Regular Expressions?

- A. nltk
- B. string
- C. re
- D. regex_tools

36. Characters with a special meaning in regular expressions are called:

- A. Tags
- B. Lemmas
- C. Tokens
- D. Metacharacters

37. A backslash (\) followed by one of a list of characters that has a special meaning is called:

- A. A set
- B. A token
- C. A special sequence
- D. A metacharacter group

38. A set of characters inside a pair of square brackets [] with a special meaning is called:

- A. A literal
- B. A special sequence
- C. A set
- D. A flag

39. Which RegEx function returns a list containing ALL matches?

- A. findall()
- B. search()
- C. split()
- D. sub()

40. Which RegEx function returns a Match object for only the FIRST occurrence of a match?

- A. split()
- B. search()
- C. sub()
- D. findall()

41. Which RegEx function returns a list where the string has been split at each match?

- A. split()
- B. sub()
- C. findall()
- D. search()

42. Which RegEx function replaces the matches with text of your choice?

- A. sub()
- B. search()
- C. split()
- D. findall()

43. If no matches are found, the findall() function returns None.

True / False

44. If no matches are found, the search() function returns the value None.

True / False

45. In the split() function, the parameter that controls the number of occurrences (splits) is called:

- A. limit
- B. n
- C. maxsplit
- D. count

46. "search" returns all matches, whereas "findall" returns only the first match in the string.

True / False

47. Regular expressions can be used to validate whether user input matches an email pattern such as xxx@xxx.com.

True / False

48. What is the expected output of the following code?

```
import re
s = "The rain in Germany"
x = re.search("^The.*Germany$", s)
print(x)
```

- A. None
- B. An empty list
- C. A match object indicating a match was found
- D. An error

49. Is there a match for the following program? (True or False?)

```
import re
s = "The rain in Poland"
x = re.search("rain", s)
print(x)
```

True / False

50. Is there a match for the following program? (True or False?)

```
import re
s = "The rain in Poland"
x = re.search("^The.*poland$", s)
print(x)
```

True / False

51. What is the expected output ("number of matches") of the following program?

```
import re
s = "The rain in Burkina Faso, Dominican Republic, Equatorial Guinea, Finland,\nIndia, China, Argentina, Bahrain, Indonesia, Bosnia and Herzegovina"
x = re.findall("in", s)
print(x)
```

- A. 8
- B. 11
- C. 13
- D. 10

52. Given the code below, what does x contain?

```
txt = "The rain in Spain"
x = re.findall("ai", txt)
print(x)
```

- A. ['ai']
- B. ['ai', 'ai']
- C. []

D. 2

53. Given the code below, what is the output?

```
txt = "The rain in Spain"  
x = re.sub("\\s", "9", txt)  
print(x)
```

- A. The9rain9in9Spain
- B. 9The9rain9in9Spain9
- C. The rain in Spain
- D. Error

Text Pre-Processing: Tokenization

54. Which of the following splits raw text into small units (tokens) for processing?

- A. Stop word removal
- B. Tokenization
- C. POS Tagging
- D. Lemmatization

55. Tokenization is a mandatory and fundamental step in both traditional NLP methods and modern Deep Learning architectures like Transformers.

True / False

56. Which of the following is NOT one of the tokenization unit types mentioned in the lecture?

- A. N-grams
- B. Words
- C. Sentences
- D. Paragraphs

57. Which NLTK function is used for word tokenization?

- A. RegexpTokenizer.split
- B. sent_tokenize
- C. word_tokenize
- D. ngrams

58. Which NLTK function is used for sentence tokenization?

- A. ngrams
- B. word_tokenize
- C. sent_tokenize
- D. pos_tag

59. Which NLTK utility is used to generate N-grams from a list of word tokens?

- A. RegexpTokenizer
- B. word_tokenize
- C. sent_tokenize
- D. ngrams

60. Which NLTK class allows tokenization based on a custom regular expression pattern?

- A. PorterStemmer

- B. RegexpTokenizer
- C. sent_tokenize
- D. word_tokenize

61. Tokenizing 'Natural Language Processing' by words would primarily produce:

- A. ['Natural Language Processing']
- B. ['Natural', 'Language', 'Processing']
- C. ['Natural-Language-Processing']
- D. ['N', 'L', 'P']

Text Pre-Processing: Stop Words

62. Stop word removal is best described as:

- A. Removing punctuation only
- B. Removing words that occur commonly across all documents in a corpus
- C. Converting words to their root form
- D. Removing the rarest words in a corpus

63. Stop word removal is one of the most used preprocessing steps across different NLP applications.

True / False

64. Typically, which categories of words are classified as stop words?

- A. Adjectives
- B. Proper nouns
- C. Numbers
- D. Articles and pronouns

65. Which Python package/module is used in the lecture to import a stop word list?

- A. nltk.tag
- B. nltk.stem
- C. nltk.corpus stopwords
- D. nltk.tokenize

66. Which of the following words would commonly appear in NLTK's English stop word list?

- A. tokenize
- B. the
- C. elephant
- D. Python

Text Pre-Processing: Stemming & Lemmatization

67. Stemming works by:

- A. Adding affixes to a word
- B. Translating a word into another language
- C. Counting word frequency
- D. Chopping off the ends of words to convert them to their stem or root

68. In many cases, the result of stemming is itself a valid dictionary word.

True / False

69. Which of the following is NOT a stemming algorithm mentioned in NLTK?
- A. Snowball Stemmer
 - B. Porter Stemmer
 - C. Lancaster Stemmer
 - D. WordNet Stemmer
70. Which converts words to their root form, where the result is always a real dictionary word?
- A. Stemming
 - B. Stop word removal
 - C. Tokenization
 - D. Lemmatization
71. Which of the following converts words to their root using a dictionary lookup (e.g., WordNet) plus the word's part of speech?
- A. Lemmatization
 - B. Stop word removal
 - C. Tokenization
 - D. Stemming
72. Which of the following converts words to their root?
- A. Lemmatization
 - B. Tokenization
 - C. Tagging
 - D. NER
73. Which approach is the best choice when you are looking for qualitative results?
- A. Stop word removal
 - B. Lemmatization
 - C. Tokenization
 - D. Stemming
74. Which approach is the best choice when you are looking for optimized speed?
- A. Stemming
 - B. NER
 - C. Lemmatization
 - D. POS Tagging
75. Lemmatization is generally harder to implement and slower than stemming.
True / False
76. Stemming uses vocabulary (corpus) and morphological analysis to reduce words to their base form.
True / False
77. Lemmatization uses rough heuristics to reduce words to a base form, with no guarantee the result is a real word.
True / False
78. Each word must have what attached to it before lemmatization occurs?
- A. A Part of Speech (POS) tag
 - B. A stop word flag

- C. An N-gram count
- D. A TF-IDF score

79. Lemmatization relies on which dictionary/lexical resource mentioned in the lecture?

- A. Brown Corpus
- B. Penn Treebank
- C. Gutenberg Corpus
- D. WordNet

Text Pre-Processing: POS Tagging

80. Which of the following gets information about each word's part of speech (verb, noun, adjective, etc.)?

- A. Stop word removal
- B. Tokenization
- C. Stemming
- D. Tagging

81. POS Tagging is also called grammatical tagging.

True / False

82. POS Tagging marks up words according to:

- A. Their length
- B. Their part of speech based on definition and context
- C. Their stem
- D. Their frequency in the corpus

83. The simplest possible tagger, which assigns the same tag to every token, is called the:

- A. Regular Expression Tagger
- B. N-Gram Tagger
- C. Brill's Tagger
- D. Default Tagger

84. The Default Tagger establishes an important baseline for tagger performance.

True / False

85. Which tagger assigns tags to tokens based on matching patterns, such as words ending in 'ed' being tagged as past participles?

- A. Brill's Tagger
- B. Unigram Tagger
- C. Regular Expression Tagger
- D. Default Tagger

86. A tagger that assigns the tag that is most likely for a token regardless of any context is called a:

- A. Bigram tagger
- B. Default tagger
- C. Unigram tagger
- D. Trigram tagger

87. A unigram tagger uses one item of context (the current token in isolation) when tagging.

True / False

88. Consider the code below. What is the type of this tagger?

```
size = int(len(brown_tagged_sents) * 0.9)
train_sents = brown_tagged_sents[:size]
test_sents = brown_tagged_sents[size:]
unigram_tagger = nltk.UnigramTagger(train_sents)
```

- A. N-Gram Tagger
- B. Default Tagger
- C. Brill's Tagger
- D. Regular Expression Tagger

89. It is good practice to train and test a tagger on the exact same data to maximize accuracy.

True / False

90. What is the typical recommended train/test split ratio mentioned in the lecture for tagger evaluation?

- A. 70% train / 30% test
- B. 90% train / 10% test
- C. 10% train / 90% test
- D. 50% train / 50% test

91. In N-Gram Tagging, if the tags of the two preceding words in addition to the current word are considered, what type of tagger is this?

- A. Unigram tagger
- B. Bigram tagger
- C. Trigram tagger
- D. Default tagger

92. 2-gram taggers are also called:

- A. Trigram taggers
- B. Unigram taggers
- C. Bigram taggers
- D. Default taggers

93. An n-gram tagger's context is the current word together with the part-of-speech tags of:

- A. Only the previous word's stem
- B. The n+1 following tokens
- C. All tokens in the sentence
- D. The n-1 preceding tokens

94. Combining a bigram tagger, a unigram tagger, and a default tagger as fallbacks is called:

- A. Combining taggers (backoff tagging)
- B. Regular expression tagging
- C. Brill's tagging
- D. Trigram tagging

95. In a combining/backoff tagger setup, if the bigram tagger cannot find a tag, the system falls back to the unigram tagger, and finally to the default tagger.

True / False

96. Which tagger is described as a 'transformational rule-based tagger' that guesses tags then goes back and fixes mistakes?

- A. Default Tagger
- B. Brill's Tagger
- C. Unigram Tagger
- D. Regular Expression Tagger

97. Brill's Tagger is a supervised learning method that requires annotated training data.

True / False

98. Unlike n-gram tagging, Brill's Tagger does not count observations but compiles a list of transformational correction rules.

True / False

99. Brill's Tagger rules are generated from a template described as:

- A. 'Delete T1 if followed by T2'
- B. 'Insert T1 before T2'
- C. 'Merge T1 and T2'
- D. 'Replace T1 with T2 in the context C'

100. Each Brill's Tagger rule is scored according to its net benefit, defined as:

- A. The total number of tags in the corpus
- B. The number of training sentences
- C. The number of incorrect tags it corrects minus the number of correct tags it incorrectly modifies
- D. The frequency of the word in the corpus

Text Pre-Processing: Corpus & Automatic Tagging

101. A large set of texts, often electronically stored and processed, used for linguistic research is called a:

- A. Vocabulary
- B. Treebank
- C. Corpus
- D. Lexicon

102. The process of adding linguistic information (such as POS tags) to a corpus is called:

- A. Normalization
- B. Annotation
- C. Vectorization
- D. Tokenization

103. The Brown Corpus was created in which year?

- A. 1945
- B. 1961
- C. 1950
- D. 1990

104. The Brown Corpus was the first million-word electronic corpus of English.

True / False

105. At which university was the Brown Corpus created?

- A. MIT
- B. Stanford University
- C. Harvard University
- D. Brown University

106. The Brown Corpus contains text from 500 sources categorized by type such as news, editorial, and government.

True / False

107. In Python 3 NLTK, the `evaluate()` function for taggers has been deprecated in favor of `accuracy(gold)`.

True / False

Text Pre-Processing: NER

108. NER stands for:

- A. Natural Entity Retrieval
- B. Natural Entity Recognition
- C. Named Element Recognition
- D. Named Entity Recognition

109. NER is also known as all of the following EXCEPT:

- A. Entity identification
- B. Entity chunking
- C. Entity extraction
- D. Part of speech tagging

110. NER seeks to locate and classify named entities into predefined categories such as person names, organizations, and locations.

True / False

111. NER is described in the lecture as a subtask of:

- A. Information extraction
- B. Text summarization
- C. Machine translation
- D. Sentiment analysis

Text Representations

112. Text Representation is best defined as:

- A. The process of translating text between languages
- B. The conversion of text into numbers (vectors) for machines to understand
- C. The process of summarizing long documents
- D. The process of removing stop words from text

113. Text representation plays an important role in defining features for machine and deep learning models.

True / False

114. In One-Hot Encoding, each word in the vocabulary is represented by a vector of size:

- A. Always 2
- B. N , where N is the total number of words in the vocabulary
- C. The length of the word in characters
- D. Always 100

115. In One-Hot Encoding, the encoding of a given word sets:

- A. The corresponding element to one and all others to zero
- B. All elements to the word's frequency
- C. All elements to one
- D. The corresponding element to the word's length

116. If a vocabulary contains 8 words, what is the size of the One-Hot vector for each word?

- A. 8
- B. 64
- C. 1
- D. 16

117. Which problem refers to all sorts of issues that arise with data in very high dimensions, as seen in One-Hot Encoding?

- A. Vanishing gradient
- B. Data leakage
- C. The curse of dimensionality
- D. Overfitting

118. For a vocabulary of 50,000 words, how much memory (in units) is needed for the full One-Hot representation according to the lecture's example?

- A. 2.5 million
- B. 100,000
- C. 2.5 billion
- D. 50,000

119. With One-Hot vectors, all words are equal distance apart, meaning there is no meaningful comparison between them.

True / False

120. One-Hot Encoding considers the frequency of a word in the document.

True / False

121. One-Hot Encoding utilizes context information between words.

True / False

122. A Bag-of-Words (BoW) model is a way of:

- A. Encrypting text data
- B. Compressing images
- C. Extracting features from text for use in modeling
- D. Translating text

123. In a Bag-of-Words model, information about the order or structure of words in the document is discarded.

True / False

124. A Bag-of-Words model involves two main things:

- A. A neural network and a loss function
- B. A vocabulary of known words and a measure of the presence of known words
- C. A grammar and a parser
- D. A corpus and a translation dictionary

125. Which one of the following text representation methods is called Term-Document (Document-Term) Matrix?

- A. Bag of Words
- B. Co-Occurrence Matrix
- C. One-Hot Encoding
- D. Occurrence Matrix

126. Which of the following is a limitation of the Bag-of-Words model related to harder modeling due to so little information in a large space?

- A. Tokenization error
- B. Overfitting
- C. Sparsity
- D. Vocabulary size only

127. Discarding word order in BoW, which causes loss of context and meaning, is referred to in the lecture as the limitation of:

- A. Vocabulary
- B. Meaning
- C. Frequency
- D. Sparsity

128. On an elementary level, the Bag-of-Words approach is implemented in practice using:

- A. One-Hot Encoder
- B. Count Vectorizer
- C. Word2Vec
- D. TF-IDF Vectorizer

129. Count Vectorizer assigns each position of a count vector to:

- A. A sentence length
- B. A character in the alphabet
- C. A random number
- D. A particular token, with its value representing the number of appearances

130. Which vectorizer is memory efficient and applies a hashing trick to encode tokens as numerical indexes instead of storing them as strings?

- A. One-Hot Encoder
- B. Count Vectorizer
- C. Hashing Vectorizer
- D. TF-IDF Vectorizer

131. TF (Term Frequency) is calculated as:

- A. $(\text{Number of times term } t \text{ appears in a document}) / (\text{Total number of terms in the document})$
- B. $\log(\text{Total documents} / \text{documents containing the term})$

- C. $(\text{Number of unique terms}) / (\text{Total documents})$
- D. $(\text{Number of documents}) / (\text{Number of documents containing term } t)$

132. IDF (Inverse Document Frequency) measures:

- A. How unique or rare a word is across documents
- B. The number of stop words in a document
- C. The length of a document
- D. How frequently a word appears in one document

133. IDF(t) is calculated as:

- A. $\text{Total number of documents} * \text{Number of documents with term } t$
- B. $\log(\text{Total number of documents} / \text{Number of documents with term } t \text{ in it})$
- C. $(\text{Number of times } t \text{ appears}) / (\text{Total terms})$
- D. $\text{Number of documents with term } t / \text{Total number of documents}$

134. A document contains 100 words, and the word 'apple' appears 5 times. What is the TF for 'apple'?

- A. 5
- B. 20
- C. 0.5
- D. 0.05

135. Given 10,000,000 documents, where 'apple' appears in 1,000 of them, what is the IDF (using log base 10) of 'apple'?

- A. 10
- B. 0.0001
- C. 1000
- D. 4

136. Given $TF = 0.05$ and $IDF = 4$, what is the TF-IDF weight for the word 'apple'?

- A. 20
- B. 4.05
- C. 0.20
- D. 0.05

137. A high weight in TF-IDF is reached by a high term frequency in the given document and a high document frequency across the whole collection.

True / False

138. N-Gram representation differs from the Count Vectorizer bag-of-words model in that it:

- A. Encodes words as one-hot vectors
- B. Removes all stop words automatically
- C. Always normalizes text to lowercase first
- D. Calculates the frequency of groups of words occurring together, instead of single words

139. An N-Gram model with groups of 2 words is called a:

- A. Quadgram
- B. Unigram
- C. Bigram
- D. Trigram

140. An N-Gram model with groups of 3 words is called a:

- A. Bigram
- B. Trigram
- C. 4-gram
- D. Unigram

141. An Occurrence Matrix has context (document, sentence, tweet, post) on one axis and what on the other axis?

- A. Timestamps
- B. Terms (vocabulary)
- C. Sentence lengths
- D. Authors

142. Each cell of an Occurrence Matrix can contain 0/1, the Frequency, or the TF-IDF score.

True / False

143. Unlike the Occurrence Matrix (rectangular), the Co-Occurrence Matrix is:

- A. A sparse one-hot matrix
- B. A square matrix
- C. Always binary
- D. A diagonal matrix

144. The Co-Occurrence Matrix is also called the:

- A. Term-Document matrix
- B. Sentence-Term matrix
- C. Document-Document matrix
- D. Term-Term matrix

145. In the 'Full Document' approach to building a Co-Occurrence Matrix, two terms are considered co-occurring if:

- A. They have the same stem
- B. They occur in the same document
- C. They occur in the same sentence only
- D. They have the same part of speech

146. In the 'Window' approach to building a Co-Occurrence Matrix, two terms are considered co-occurring if:

- A. They are both stop words
- B. They appear in the same paragraph only
- C. They occur within the same context window of a given size
- D. They have identical TF-IDF scores

147. The Window approach for Co-Occurrence captures both syntactic and semantic information.

True / False

148. Co-occurrence matrices are easy to incorporate new words or documents into without significant rework.

True / False

149. Co-occurrence matrices increase in size as the vocabulary grows and require a lot of storage.

True / False

150. The phrase 'A word can be identified by the company it keeps' is associated with which concept in the lecture?

- A. Stop word removal
- B. Distributional Representations
- C. One-Hot Encoding
- D. Stemming

PART B — Original Mock Exam & Quiz Questions (Verbatim)

These are reproduced exactly as they appeared in your uploaded files, including any questions that are repeated between the two mock-exam attempts.

Mid-Term Mock Exam (Attempt 1)

151. Consider the sentence "I can go". The word embeddings are: I = [0.1, 0.4, 0.4] can = [0.9, 0.9, 0.2] go = [0.5, 0.5, 0.6] Find the sentence embedding by two methods: Sum and Avg.

(Short-answer / fill-in question — see answer key)

152. Which one of the following text representation methods is called Term-Document (Document-Term) Matrix?

- A. One-Hot Encoding.
- B. None of these choices.
- C. All of these choices.
- D. Co-Occurrence Matrix
- E. Bag of Words.
- F. Occurrence Matrix.

153. Match the NLP application term with its definition:

- **Sentiment analysis** → is the process of analyzing text to determine if the emotional tone of the message is positive, negative, or neutral.
- **Predictive text** → is an input technology that facilitates typing on a device by suggesting words the user may wish to insert in a text field.
- **Paraphrasing** → is the process that helps students and professionals rewrite, edit, and change the tone of their text to improve clarity.
- **Social media monitoring** → is the process of identifying and determining what is being said about a brand, individual or product through different social and online channels.
- **Email filtering** → is often used to detect spam, viruses, and malware before it reaches a user.
- **Chatbot** → is a computer program that simulates human conversation with an end user.
- **Automatic summarization in NLP** → is the process of reducing the text size and create a summary of the text data.

154. Match the NLP application term with its definition: (second instance, different order)

- **Predictive text** → is an input technology that facilitates typing on a device by suggesting words the user may wish to insert in a text field.
- **Social media monitoring** → is the process of identifying and determining what is being said about a brand, individual or product through different social and online channels.
- **Automatic summarization in NLP** → is the process of reducing the text size and create a summary of the text data.
- **Paraphrasing** → is the process that helps students and professionals rewrite, edit, and change the tone of their text to improve clarity.
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- **Email filtering** → is often used to detect spam, viruses, and malware before it reaches a user.
- **Chatbot** → is a computer program that simulates human conversation with an end user.

155. Which of the following get information about each word's part of speech (verb, noun, adjective, etc.)?

- A. Tagging
- B. Stemming
- C. Stop word removal
- D. None of these choices.
- E. Lemmatization
- F. Tokenization
- G. NER

156. Is there a match of the following program? (True or False?)

```
import re
s = "The rain in Germany"
x = re.search("^The.*Germany$", s)
print (x)
True / False
```

157. Predictive text is often used to detect spam, viruses, and malware before it reaches a user.

True / False

158. Match the model with what it predicts:

- **Continuous Bag-of-Words model** → predicting the word given its context
- **Continuous Skip-gram model** → predicting the context given a word

159. What is the expected output of the following code?

```
>>> model.most_similar(positive=['king', 'woman'], negative=['man'], topn = 1)
```

- A. All of these choices.
- B. None of these choices.
- C. 'queen'
- D. 'queens'
- E. 'prince'

160. Sentiment analysis is the process of reducing the text size and creating a summary of the text data.

True / False

161. Using the word embedding to complete the missing in the following graph, by finding A and B (graph shows vectors: A → man (analogous to king), woman → queen, and B is the vector from 'man' to 'king' applied at the 'woman'/'queen' position):

(Short-answer / fill-in question — see answer key)

162. Sentiment analysis is the process of reducing the text size and creating a summary of the text data. (repeated)

True / False

163. Is there a match of the following program? (True or False?)

```
import re
s = "The rain in Poland"
x = re.search("rain", s)
print (x)
```

True / False

164. Which of the following stages of language processing involve understanding the patterns present in the sound and speeches?

- A. Syntactic
- B. Phonetical and Phonological
- C. None of these choices.
- D. Lexical
- E. All of these choices.
- F. Semantic
- G. Morphological

165. The Turing test measures the performance of an intelligent machine against that of a human being.

True / False

166. Is there a match of the following program? (True or False?)

```
import re
s = "The rain in Poland"
x = re.search("r.*n", s)
print (x)
```

True / False

Mid-Term Mock Exam (Attempt 2)

167. Consider the sentence "I can go". The word embeddings are: I = [0.1, 0.4, 0.4] can = [0.9, 0.9, 0.2] go = [0.5, 0.5, 0.6] Find the sentence embedding by two methods: Sum and Avg.

(Short-answer / fill-in question — see answer key)

168. Which one of the following text representation methods is called Term-Document (Document-Term) Matrix?

- A. One-Hot Encoding.
- B. None of these choices.
- C. All of these choices.
- D. Co-Occurrence Matrix
- E. Bag of Words.
- F. Occurrence Matrix.

169. Match the NLP application term with its definition:

- **Sentiment analysis** → is the process of analyzing text to determine if the emotional tone of the message is positive, negative, or neutral.
- **Predictive text** → is an input technology that facilitates typing on a device by suggesting words the user may wish to insert in a text field.
- **Paraphrasing** → is the process that helps students and professionals rewrite, edit, and change the tone of their text to improve clarity.
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- **Email filtering** → is often used to detect spam, viruses, and malware before it reaches a user.
- **Chatbot** → is a computer program that simulates human conversation with an end user.

- **Automatic summarization in NLP** → is the process of reducing the text size and create a summary of the text data.

170. Match the NLP application term with its definition: (second instance, different order)

- **Predictive text** → is an input technology that facilitates typing on a device by suggesting words the user may wish to insert in a text field.
- **Social media monitoring** → is the process of identifying and determining what is being said about a brand, individual or product through different social and online channels.
- **Automatic summarization in NLP** → is the process of reducing the text size and create a summary of the text data.
- **Paraphrasing** → is the process that helps students and professionals rewrite, edit, and change the tone of their text to improve clarity.
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- C. Stop word removal
- D. None of these choices.
- E. Lemmatization
- F. Tokenization
- G. NER

172. Is there a match of the following program? (True or False?)

```
import re
s = "The rain in Germany"
x = re.search("^The.*Germany$", s)
print (x)
True / False
```

173. Predictive text is often used to detect spam, viruses, and malware before it reaches a user.

True / False

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- **Continuous Skip-gram model** → predicting the context given a word

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```
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```

- A. All of these choices.
- B. None of these choices.
- C. 'queen'
- D. 'queens'
- E. 'prince'

176. Sentiment analysis is the process of reducing the text size and creating a summary of the text data.

True / False

177. Using the word embedding to complete the missing in the following graph, by finding A and B (graph shows vectors: $A \rightarrow \text{man}$ (analogous to king), $\text{woman} \rightarrow \text{queen}$, and B is the vector from 'man' to 'king' applied at the 'woman'/'queen' position):

(Short-answer / fill-in question — see answer key)

178. Sentiment analysis is the process of reducing the text size and creating a summary of the text data. (repeated)

True / False

179. Is there a match of the following program? (True or False?)

```
import re
s = "The rain in Poland"
x = re.search("rain", s)
print (x)
```

True / False

180. Which of the following stages of language processing involve understanding the patterns present in the sound and speeches?

- A. Syntactic
- B. Phonetical and Phonological
- C. None of these choices.
- D. Lexical
- E. All of these choices.
- F. Semantic
- G. Morphological

181. The Turing test measures the performance of an intelligent machine against that of a human being.

True / False

182. Is there a match of the following program? (True or False?)

```
import re
s = "The rain in Poland"
x = re.search("r.*n", s)
print (x)
```

True / False

Quiz 1

183. NLP is a subfield of:

- A. linguistics
- B. None of these choices.
- C. computer science
- D. All of these choices.
- E. artificial intelligence

184. Which of the following stages of language processing involve understanding the part of speech?

- A. Lexical
- B. Pragmatic
- C. All of these choices.
- D. Discourse
- E. Phonetical and Phonological
- F. Syntactic
- G. Morphological
- H. Semantic
- I. None of these choices.

185. Consider the Turing Test. Match each role:

- **The human questioner is the** → Interrogator.
- **The computer respondent is the** → AI.
- **The human respondent is the** → Expert.

186. Consider the following graph (Arabic text examples with English translations) — match each item (i, ii, iii, iv) to its NLP task:

- **(iv)** → Paraphrasing.
- **(ii)** → Machine translation.
- **(iii)** → Question generation.
- **(i)** → Title generation.

Quiz 2

187. Which of the following converts words to their root?

- A. None of these choices.
- B. NER
- C. Tokenization
- D. Stop word removal
- E. Lemmatization
- F. Tagging
- G. Named Entity Recognition

188. What is the expected output "number of matches" of the following program?

```
import re
s = "The rain in Burkina Faso, Dominican Republic, Equatorial Guinea,
Finland,\nIndia, China, Argentina, Bahrain, Indonesia, Bosnia and Herzegovina"
x = re.findall("in", s)
print (x)
```

- A. 8
- B. 4
- C. 13
- D. 11
- E. 3
- F. 0
- G. 10
- H. None of these choices.

- I. 1
- J. 5
- K. 6
- L. 2
- M. 12
- N. 7
- O. 9

189. Match:

- **Lemmatization** → is the best choice when you are looking for qualitative results.
- **Stemming** → is the best choice when you are looking for optimized speed.

190. Stop word removal is one of the most used ----- steps across different NLP applications.

- A. Postprocessing.
- B. Preprocessing.
- C. Main processing.
- D. All of these choices.
- E. None of these choices.

191. Is there a match of the following program? (True or False?)

```
import re
s = "The rain in Poland"
x = re.search("^The.*poland$", s)
print (x)
True / False
```

192. "search" returns all matches, whereas "findall" returns the first match in the string.

True / False

Quiz 3

193. Text preprocessing involves the cleaning of text before the application of NLP techniques.

True / False

194. Which of the following splits raw text into small units (tokens) for processing?

- A. Tokenization
- B. None of these choices.
- C. Stop word removal
- D. Stemming
- E. Lemmatization
- F. Tagging

195. In N-Gram Tagging, if the tags of the two preceding words in addition to the current word is considered, what is the type of the tagger?

- A. Trigram tagger
- B. Bigram tagger
- C. All of these choices.
- D. Unigram tagger
- E. None of these choices.

Quiz 4

196. Consider the following code. What is the type of the tagger?

```
>>> size = int(len(brown_tagged_sents) * 0.9)
>>> size
4160
>>> train_sents = brown_tagged_sents[:size]
>>> test_sents = brown_tagged_sents[size:]
>>> unigram_tagger = nltk.UnigramTagger(train_sents)
>>> unigram_tagger.evaluate(test_sents)
0.811721...
```

- A. Default Tagger
- B. Regular Expression Tagger
- C. N-Gram Tagger
- D. Combining Tagger
- E. Brill's Tagger
- F. All of these choices.
- G. None of these choices.

197. What is the type of the following tagger? (table shows Phrase/Unigram/Rule1/Rule2/Output/Gold tags for: to increase grants to states for vocational rehabilitation)

- A. Default Tagger
- B. Regular Expression Tagger
- C. N-Gram Tagger
- D. Combining Tagger
- E. Brill's Tagger
- F. None of these choices.

PART C — Lecture 5: New Practice Questions

Covers the uploaded Lecture 5 slide deck (Word2vec: CBOW, Skip-gram, Negative Sampling, Training Process, Gensim) and the uploaded Sentence Embeddings / RNN / LSTM slide images.

Sentence Embeddings (Introduction)

198. Sentence embedding is best defined as:

- A. A set of techniques in NLP where sentences are mapped to vectors of real numbers
- B. A method to translate sentences between languages
- C. A technique to remove stop words from a sentence
- D. A way to tag each word in a sentence with its part of speech

199. Popular sentence embeddings are based on the hidden layer outputs of which type of model?

- A. Decision tree models
- B. Transformer models like BERT (SBERT)
- C. Naïve Bayes models
- D. Regular Expression models

200. An alternative direction to transformer-based sentence embeddings is to:

- A. Remove all stop words from the sentence
- B. Aggregate word embeddings, such as those returned by Word2vec, into sentence embeddings
- C. Translate the sentence into another language first
- D. Tag every word with its part of speech

201. The most straightforward approach to aggregating word embeddings into a sentence embedding is to compute the:

- A. Maximum of word vectors
- B. Sum or average of word vectors (CBOW)
- C. Product of word vectors
- D. Median of word vectors

202. "CBOW" in the context of sentence embeddings stands for:

- A. Context-Based Order of Words
- B. Continuous Bag-of-Words
- C. Combined Bag of Word-vectors
- D. Contextual Bag of Weights

203. Given $\text{vec}(\text{"dog"}) = (0.2, -0.3, 1.5)$, $\text{vec}(\text{"bites"}) = (0.5, 1.0, -0.4)$, and $\text{vec}(\text{"man"}) = (-0.1, 2.3, -1.5)$, what is the Sum sentence embedding for "dog bites man"?

- A. (0.6, 3.0, -0.4)
- B. (0.2, 1.0, -0.133)
- C. (0.8, 3.0, -0.4)
- D. (0.6, 1.0, -0.4)

204. Given the same vectors as above, what is the Avg sentence embedding for "dog bites man"?

- A. (0.6, 3.0, -0.4)
- B. (0.2, 1.0, -0.133)
- C. (0.3, 1.0, -0.13)

D. (0.18, 0.07, -0.5)

205. According to the lecture, simply summing or averaging word vectors clearly has a lot missing because it cannot distinguish:

- A. "man bites dog" from "dog bites man"
- B. Nouns from verbs
- C. Stop words from content words
- D. Tokens from lemmas

206. The sentence "the quick, brown fox jumps over the lazy dog" producing the same sum/average embedding as its word-shuffled version illustrates that simple aggregation approaches ignore:

- A. Spelling
- B. Word order
- C. Capitalization
- D. Punctuation only

Recurrent Neural Network (RNN) & LSTM

207. The main motivation behind the Recurrent Neural Network (RNN) architecture is the need for:

- A. Sequential modeling
- B. Image classification
- C. Dimensionality reduction
- D. Stop word removal

208. Which of the following is NOT listed in the lecture as an example of sequence data?

- A. Speech recognition
- B. Machine translation
- C. Named entity recognition
- D. One-hot encoding

209. A recurrent neural network (RNN) is best described as a neural network that is specialized for processing:

- A. A single fixed-size vector
- B. A sequence of data $x(t) = x(1), \dots, x(T)$
- C. Only images
- D. Only tabular data

210. RNNs are called "recurrent" because they:

- A. Use a different architecture for every input
- B. Perform the same task for every element of a sequence, with the output depending on previous computations
- C. Repeat training multiple times on the same dataset
- D. Only work on recurring (duplicate) words

211. Another way to think about RNNs, according to the lecture, is that they have a:

- A. "Memory" which captures information about what has been calculated so far
- B. "Filter" which removes irrelevant tokens
- C. "Translator" which converts text to vectors
- D. "Classifier" which only outputs binary results

212. In the RNN diagram processing "What time is it?", after processing "What" and producing output O1, what is fed into the network along with the next word "time" to produce O2?

- A. Nothing; each word is processed independently
- B. The output O1 (carrying memory of the previous word)
- C. Only the word "is"
- D. The entire vocabulary

213. In the unrolled RNN diagram, $h(0)$ together with $X(1)$ serves as the input for the next step. This illustrates that the RNN:

- A. Discards all previous information at every step
- B. Keeps remembering the context while training
- C. Cannot process more than two inputs
- D. Requires the entire sequence to be known before starting

214. In the "one-to-many" RNN architecture, the network deals with:

- A. A sequence of inputs producing a single output
- B. A fixed size of information as input that gives a sequence of data as output
- C. A sequence of inputs producing a sequence of outputs
- D. No input or output at all

215. Image captioning, where a single image produces a sentence of words, is an example of which RNN architecture?

- A. Many-to-one
- B. Many-to-many
- C. One-to-many
- D. One-to-one

216. Sentiment analysis, where a sentence is classified as expressing positive or negative sentiment, is an example of which RNN architecture?

- A. One-to-many
- B. Many-to-one
- C. Many-to-many
- D. One-to-one

217. Machine translation, where the RNN reads a sentence in English and outputs the equivalent sentence in French, is an example of which RNN architecture?

- A. One-to-many
- B. Many-to-one
- C. Many-to-many
- D. One-to-one

218. A gradient, as defined in the lecture, is best thought of as:

- A. A measure of how much the output of a function changes if the inputs change a little bit
- B. The total number of layers in a network
- C. The size of the training dataset
- D. The number of words in the vocabulary

219. According to the lecture, a gradient can also be thought of as:

- A. The vocabulary size

- B. The slope of a function
- C. The number of epochs
- D. The learning rate

220. According to the lecture, the higher the gradient, the:

- A. Slower the model learns
- B. Steeper the slope and the faster a model can learn
- C. Smaller the vocabulary becomes
- D. Less accurate the model becomes

221. According to the lecture, if the slope (gradient) is zero, the model:

- A. Learns instantly
- B. Stops learning
- C. Becomes more accurate
- D. Doubles its vocabulary

222. Exploding gradients occur when the algorithm:

- A. Assigns a stupidly high importance to the weights without much reason
- B. Assigns zero importance to all weights
- C. Removes all stop words from the training data
- D. Runs out of training data

223. According to the lecture, the exploding gradients problem can be easily solved by:

- A. Increasing the vocabulary size
- B. Truncating or squashing the gradients
- C. Removing the hidden layer entirely
- D. Adding more stop words

224. Vanishing gradients occur when:

- A. The values of a gradient are too small, and the model stops learning or takes too long
- B. The values of a gradient are too large, causing instability
- C. The vocabulary is too small
- D. The training data has too many duplicates

225. According to the lecture, the vanishing gradients problem was fortunately solved through the concept of:

- A. One-Hot Encoding
- B. TF-IDF
- C. LSTM
- D. Stop word removal

226. Long Short-Term Memory networks (LSTMs) are best described as:

- A. A replacement for tokenization
- B. An extension for RNNs which basically extends the memory
- C. A type of stop word list
- D. A stemming algorithm

227. LSTM is well suited to learn from important experiences that have:

- A. Very short time lags in between
- B. Very long time lags in between

- C. No time lag at all
- D. Only a single time step

228. In a Long Short-Term Memory cell, how many gates are there?

- A. Two
- B. Three
- C. Four
- D. Five

229. Which of the following is NOT one of the three gates in an LSTM cell, according to the lecture?

- A. Input gate
- B. Forget gate
- C. Output gate
- D. Update gate

230. In an LSTM cell, which gate determines whether to delete information because it isn't important?

- A. Input gate
- B. Forget gate
- C. Output gate
- D. Reset gate

231. In an LSTM cell, which gate determines whether to let new input in?

- A. Forget gate
- B. Output gate
- C. Input gate
- D. Update gate

232. In an LSTM cell, which gate determines whether to let information impact the output at the current timestep?

- A. Input gate
- B. Forget gate
- C. Output gate
- D. Reset gate

233. The GRU (Gated Recurrent Unit) architecture, as contrasted with LSTM in the lecture's diagram, uses a "reset gate" and which other gate?

- A. Forget gate
- B. Output gate
- C. Update gate
- D. Input gate

Word2vec

234. Word2vec uses a neural network model to learn word associations from:

- A. A single sentence
- B. A large corpus of text
- C. A dictionary of synonyms only
- D. A list of stop words

235. As the name implies, word2vec represents each distinct word with a particular list of numbers called a:

- A. Tag
- B. Token
- C. Vector
- D. Lemma

236. According to the lecture, which mathematical function can indicate the level of semantic similarity between word vectors in word2vec?

- A. Euclidean distance only
- B. Cosine similarity
- C. Edit distance
- D. Jaccard index

237. Word2vec was introduced by:

- A. Geoffrey Hinton
- B. Tomas Mikolov
- C. Yoshua Bengio
- D. Andrew Ng

238. The key idea behind word2vec, as stated in the lecture, is:

- A. Count, don't predict
- B. Don't count, predict
- C. Translate, then count
- D. Tag, then count

239. Word2vec predicts the surrounding words of every word, and as a result it is:

- A. Slower and cannot incorporate new words
- B. Faster and can easily incorporate a new sentence/document or add a word to the vocabulary
- C. Only usable for image data
- D. Unable to encode word meaning spatially

240. According to the lecture, word2vec is best described as:

- A. A Neural Network approach, but NOT Deep Learning
- B. A Deep Learning approach only
- C. A rule-based tagging approach
- D. A purely statistical n-gram approach with no neural network

241. Word2vec proposed two architectures. What are they?

- A. CBOW and TF-IDF
- B. Continuous Bag-of-Words model and Continuous Skip-gram model
- C. One-Hot Encoding and Bag-of-Words
- D. LSTM and GRU

242. In the Continuous Bag-of-Words (CBOW) architecture of word2vec, the model:

- A. Predicts the context given a word
- B. Predicts the word given its context
- C. Predicts the part of speech of a word
- D. Predicts the sentiment of a sentence

243. In the Continuous Skip-gram architecture of word2vec, the model:

- A. Predicts the word given its context
- B. Predicts the context given a word
- C. Removes the context words entirely
- D. Predicts the sentence embedding directly

244. In the CBOW architecture diagram, the inputs $w(t-2)$, $w(t-1)$, $w(t+1)$, and $w(t+2)$ are summed/projected to predict:

- A. The center word $w(t)$
- B. The next sentence
- C. The document topic
- D. The part-of-speech tag

245. In the Skip-gram architecture diagram, the center word $w(t)$ is used as input to predict:

- A. A single fixed output class
- B. The surrounding context words $w(t-2)$, $w(t-1)$, $w(t+1)$, $w(t+2)$
- C. The part-of-speech tag of $w(t)$
- D. The stemmed form of $w(t)$

246. The famous word2vec analogy example "king" - "man" + "woman" results in a vector whose closest word (from the embeddings collection) is:

- A. "prince"
- B. "queen"
- C. "princess"
- D. "royalty"

247. According to the lecture, the resulting vector from "king - man + woman" exactly equals "queen."
True / False

248. The "Country and Capital Vectors Projected by PCA" plot shown in the lecture is used to illustrate that word2vec embeddings:

- A. Cannot capture any relationships between words
- B. Capture relationships such as country-to-capital-city in a roughly consistent spatial direction
- C. Only work for English words
- D. Require manual labeling of every country

249. Language modeling (LM), as defined in the lecture, is the use of statistical and probabilistic techniques to determine:

- A. The part of speech of a word
- B. The probability of a given sequence of words occurring in a sentence
- C. The stop words in a sentence
- D. The stem of a word

250. The next-word prediction feature of a smartphone keyboard is given in the lecture as an example application of:

- A. Named Entity Recognition
- B. Language modeling
- C. Stop word removal
- D. Lemmatization

- 251.** In practice, a language model does not output only one predicted word; instead, it outputs:
- A. A probability score for all the words in its vocabulary
 - B. A single random word
 - C. Only nouns
 - D. The most frequent word in the corpus overall
- 252.** Word embeddings are learned, according to the lecture, by looking at:
- A. Which other words a word tends to appear next to (using a sliding window over text)
 - B. The alphabetical order of words
 - C. The length of each word
 - D. The number of vowels in each word
- 253.** In the CBOW variant described in the lecture, instead of only looking at words before the target word, the model can also look at:
- A. Words after the target word
 - B. Words in a completely different sentence
 - C. Only punctuation marks
 - D. Only stop words
- 254.** In the Skip-gram diagram described in the lecture, the word in the green slot represents:
- A. A randomly chosen negative sample
 - B. The input word, with each pink box being a possible output
 - C. The output word only
 - D. The stop word to be removed
- 255.** According to the lecture, if all training examples for the word2vec prediction task are positive (target = 1), the risk is that the model:
- A. Always returns 1, achieving 100% accuracy but learning nothing and generating garbage embeddings
 - B. Crashes immediately
 - C. Becomes too slow to train
 - D. Cannot represent any word as a vector
- 256.** To address the problem of an all-positive-example dataset, the lecture introduces:
- A. Stop word removal
 - B. Negative sampling (samples of words that are not neighbors, for which the model must output 0)
 - C. Lemmatization
 - D. POS tagging
- 257.** Skip-gram with Negative Sampling is commonly abbreviated as:
- A. SGD
 - B. SGNS
 - C. CBOW
 - D. GRU
- 258.** At the start of word2vec training, two matrices are created. What are they?
- A. Embedding matrix and Context matrix
 - B. Train matrix and Test matrix
 - C. Sparse matrix and Dense matrix

D. Forget matrix and Output matrix

259. In the Embedding and Context matrices used in word2vec training, the first dimension corresponds to vocabulary size, and the second dimension corresponds to:

- A. The number of documents
- B. The embedding size (e.g., 300)
- C. The number of negative samples only
- D. The number of sentences in the corpus

260. At the start of the word2vec training process, the Embedding and Context matrices are initialized with:

- A. All zeros
- B. All ones
- C. Random values
- D. Pretrained GloVe vectors only

261. In each word2vec training step, the model takes one positive example and its associated:

- A. Stop words
- B. Negative examples
- C. POS tags
- D. Lemmas

262. In the word2vec training process, after looking up embeddings for the input word (from the Embedding matrix) and the context words (from the Context matrix), the next step is to compute the:

- A. Edit distance between the words
- B. Dot product of the input embedding with each of the context embeddings
- C. Sum of all embeddings in the vocabulary
- D. TF-IDF score of each word

263. To turn the dot-product similarity scores into values between zero and one (probability-like), word2vec training uses the:

- A. Softmax of the entire vocabulary only
- B. Sigmoid function
- C. ReLU function
- D. Tanh of the gradient

264. After computing an error score during word2vec training, this score is used to:

- A. Discard the current training example permanently
- B. Adjust the embeddings of the input and context words so future predictions are closer to the target scores
- C. Increase the vocabulary size
- D. Remove the word from the corpus

265. After training completes, what does word2vec typically discard, keeping only the Embedding matrix as the pre-trained embeddings?

- A. The training corpus
- B. The Context matrix
- C. The vocabulary list
- D. The negative samples list

266. According to the lecture, two key hyperparameters in the word2vec training process are:

- A. Learning rate and batch size only
- B. Window size and the number of negative samples
- C. Number of layers and activation function
- D. Stop word list size and corpus size

267. In the Gensim example shown in the lecture, which line of code trains a Word2Vec model on the first 10,000 sentences of the Brown corpus?

```
>>> from nltk.corpus import brown
>>> train_set = brown.sents()[ :10000]
>>> model = gensim.models.Word2Vec(train_set)
```

- A. model = gensim.models.Word2Vec(train_set)
- B. model = nltk.Word2Vec(train_set)
- C. model = gensim.train(train_set)
- D. model = Word2Vec.fit(train_set)

268. In the Gensim example, which function call is used to save a trained Word2Vec model to disk under the name 'brown.embedding'?

```
>>> model.save('brown.embedding')
>>> new_model = gensim.models.Word2Vec.load('brown.embedding')
```

- A. model.export('brown.embedding')
- B. model.save('brown.embedding')
- C. model.write('brown.embedding')
- D. model.dump('brown.embedding')

269. According to the lecture, the full pre-trained word2vec model is trained on text from which source?

- A. Wikipedia only
- B. 100 billion words from the Google News Dataset
- C. The Brown Corpus only
- D. Twitter posts only

270. In the Gensim pre-trained model example, calling model.most_similar(positive=['university'], topn=3) is expected to return words such as:

```
>>> model.most_similar(positive=['university'], topn = 3)
[('universities', 0.70039...), ('faculty', 0.67809...), ('undergraduate',
0.65870...)]
```

- A. 'universities', 'faculty', 'undergraduate'
- B. 'queen', 'king', 'man'
- C. 'the', 'a', 'an'
- D. 'rain', 'spain', 'germany'

271. According to the lecture's "Next Lecture" preview, which of the following topics is mentioned as coming up after word2vec?

- A. Sent2vec and Doc2vec
- B. Regular expressions
- C. Stop word removal
- D. The Turing Test

PART D — Original Quiz 5 & Quiz 6 Questions (Verbatim)

These are reproduced exactly as they appeared in your uploaded Quiz 5 and Quiz 6 attempt-review files.

Quiz 5

272. Consider the following code (shown below). What is the size of the testing dataset, if the size of the training dataset is 4160?

```
>>> size = int(len(brown_tagged_sents) * 0.9)
>>> size
4160
>>> train_sents = brown_tagged_sents[:size]
>>> test_sents = brown_tagged_sents[size:]
>>> unigram_tagger = nltk.UnigramTagger(train_sents)
>>> unigram_tagger.evaluate(test_sents)
0.811721...
```

- A. 462
- B. 464
- C. 463
- D. 4160
- E. 4621
- F. 4622
- G. 4623
- H. All of these choices.
- I. None of these choices.

273. Consider the following code (shown below). What is the size in percentage (%) of the training dataset and testing dataset?

```
>>> size = int(len(brown_tagged_sents) * 0.9)
>>> size
4160
>>> train_sents = brown_tagged_sents[size:]
>>> test_sents = brown_tagged_sents[:size]
>>> unigram_tagger = nltk.UnigramTagger(train_sents)
>>> unigram_tagger.evaluate(test_sents)
0.811721...
```

- A. 90% for training and 10% for testing
- B. 90% for testing and 10% for training
- C. $\approx 81.17\%$ for training and $\approx 18.83\%$ for testing
- D. None of these choices.

Quiz 6

274. Match each diagram to its model name: (i) a memory cell diagram with Output Gate, Forget Gate, Memory Cell, and Input Gate; (ii) a feed-forward network diagram with crossed connections and no

recurrence (inputs $x_1, x_2 \rightarrow$ outputs y_1, y_2); (iii) a recurrent network diagram with "store" feedback connections (inputs $x_1, x_2 \rightarrow$ outputs y_1, y_2).

(i) $\rightarrow$ LSTM

(ii) $\rightarrow$ Feed Forward Network

(iii) $\rightarrow$ RNN

275. The output of the RNN depends only on the present inputs.

True / False

276. LSTM has a memory cell with several control gates.

True / False

Answer Key — Part A

Lecture 1: NLP Overview

1. A 2. True 3. D 4. True 5. C 6. B
7. A 8. True 9. True 10. C 11. C 12. C
13. A 14. C 15. D 16. D 17. D 18. D
19. A 20. B 21. D 22. True 23. C 24. D
25. False 26. False 27. A 28. A 29. A 30. D

Text Pre-Processing: Introduction & Regex

31. C 32. True 33. False 34. B 35. C 36. D
37. C 38. C 39. A 40. B 41. A 42. A
43. False 44. True 45. C 46. False 47. True 48. C
49. True 50. False 51. D 52. B 53. A

Text Pre-Processing: Tokenization

54. B 55. True 56. D 57. C 58. C 59. D
60. B 61. B

Text Pre-Processing: Stop Words

62. B 63. True 64. D 65. C 66. B

Text Pre-Processing: Stemming & Lemmatization

67. D 68. False 69. D 70. D 71. A 72. A
73. B 74. A 75. True 76. False 77. False 78. A
79. D

Text Pre-Processing: POS Tagging

80. D 81. True 82. B 83. D 84. True 85. C
86. C 87. True 88. A 89. False 90. B 91. C
92. C 93. D 94. A 95. True 96. B 97. True
98. True 99. D 100. C

Text Pre-Processing: Corpus & Automatic Tagging

101. C 102. B 103. B 104. True 105. D 106. True
107. True

Text Pre-Processing: NER

108. D 109. D 110. True 111. A

Text Representations

112. B 113. True 114. B 115. A 116. A 117. C
118. C 119. True 120. False 121. False 122. C 123. True
124. B 125. D 126. C 127. B 128. B 129. D
130. C 131. A 132. A 133. B 134. D 135. D
136. C 137. False 138. D 139. C 140. B 141. B
142. True 143. B 144. D 145. B 146. C 147. True
148. False 149. True 150. B

Answer Key — Part B (Mock Exam & Quizzes)

Mid-Term Mock Exam (Attempt 1)

151. Sum: $\text{vec}(\text{"I can go"}) = (1.5, 1.8, 1.2)$

Avg: $\text{vec}(\text{"I can go"}) = (0.5, 0.6, 0.4)$

152. F — Occurrence Matrix.

153. Match (see pairs below):

- **Sentiment analysis** → is the process of analyzing text to determine if the emotional tone of the message is positive, negative, or neutral.
- **Predictive text** → is an input technology that facilitates typing on a device by suggesting words the user may wish to insert in a text field.
- **Paraphrasing** → is the process that helps students and professionals rewrite, edit, and change the tone of their text to improve clarity.
- **Social media monitoring** → is the process of identifying and determining what is being said about a brand, individual or product through different social and online channels.
- **Email filtering** → is often used to detect spam, viruses, and malware before it reaches a user.
- **Chatbot** → is a computer program that simulates human conversation with an end user.
- **Automatic summarization in NLP** → is the process of reducing the text size and create a summary of the text data.

154. Match (see pairs below):

- **Predictive text** → is an input technology that facilitates typing on a device by suggesting words the user may wish to insert in a text field.
- **Social media monitoring** → is the process of identifying and determining what is being said about a brand, individual or product through different social and online channels.
- **Automatic summarization in NLP** → is the process of reducing the text size and create a summary of the text data.
- **Paraphrasing** → is the process that helps students and professionals rewrite, edit, and change the tone of their text to improve clarity.
- **Sentiment analysis** → is the process of analyzing text to determine if the emotional tone of the message is positive, negative, or neutral.
- **Email filtering** → is often used to detect spam, viruses, and malware before it reaches a user.
- **Chatbot** → is a computer program that simulates human conversation with an end user.

155. A — Tagging

156. True

157. False

158. Match (see pairs below):

- **Continuous Bag-of-Words model** → predicting the word given its context
- **Continuous Skip-gram model** → predicting the context given a word

159. C — 'queen'

160. False

161. B → man, A → king

162. False

163. True

164. B — Phonetical and Phonological
165. True
166. True

Mid-Term Mock Exam (Attempt 2)

167. Sum: $\text{vec}(\text{"I can go"}) = (1.5, 1.8, 1.2)$

Avg: $\text{vec}(\text{"I can go"}) = (0.5, 0.6, 0.4)$

168. F — Occurrence Matrix.

169. Match (see pairs below):

- **Sentiment analysis** → is the process of analyzing text to determine if the emotional tone of the message is positive, negative, or neutral.
- **Predictive text** → is an input technology that facilitates typing on a device by suggesting words the user may wish to insert in a text field.
- **Paraphrasing** → is the process that helps students and professionals rewrite, edit, and change the tone of their text to improve clarity.
- **Social media monitoring** → is the process of identifying and determining what is being said about a brand, individual or product through different social and online channels.
- **Email filtering** → is often used to detect spam, viruses, and malware before it reaches a user.
- **Chatbot** → is a computer program that simulates human conversation with an end user.
- **Automatic summarization in NLP** → is the process of reducing the text size and create a summary of the text data.

170. Match (see pairs below):

- **Predictive text** → is an input technology that facilitates typing on a device by suggesting words the user may wish to insert in a text field.
- **Social media monitoring** → is the process of identifying and determining what is being said about a brand, individual or product through different social and online channels.
- **Automatic summarization in NLP** → is the process of reducing the text size and create a summary of the text data.
- **Paraphrasing** → is the process that helps students and professionals rewrite, edit, and change the tone of their text to improve clarity.
- **Sentiment analysis** → is the process of analyzing text to determine if the emotional tone of the message is positive, negative, or neutral.
- **Email filtering** → is often used to detect spam, viruses, and malware before it reaches a user.
- **Chatbot** → is a computer program that simulates human conversation with an end user.

171. A — Tagging

172. True

173. False

174. Match (see pairs below):

- **Continuous Bag-of-Words model** → predicting the word given its context
- **Continuous Skip-gram model** → predicting the context given a word

175. C — 'queen'

176. False

177. B → man, A → king

178. False
179. True
180. B — Phonetical and Phonological
181. True
182. True

Quiz 1

183. D — All of these choices.
184. A — Lexical
185. Match (see pairs below):
 - **The human questioner is the** → Interrogator.
 - **The computer respondent is the** → AI.
 - **The human respondent is the** → Expert.**186.** Match (see pairs below):
 - **(iv)** → Paraphrasing.
 - **(ii)** → Machine translation.
 - **(iii)** → Question generation.
 - **(i)** → Title generation.

Quiz 2

187. E — Lemmatization
188. G — 10
189. Match (see pairs below):
 - **Lemmatization** → is the best choice when you are looking for qualitative results.
 - **Stemming** → is the best choice when you are looking for optimized speed.

190. B — Preprocessing.
191. False
192. False

Quiz 3

193. True
194. A — Tokenization
195. A — Trigram tagger

Quiz 4

196. C — N-Gram Tagger
197. E — Brill's Tagger

Answer Key — Part C (Lecture 5)

Sentence Embeddings (Introduction)

198. A 199. B 200. B 201. B 202. B 203. A
204. B 205. A 206. B

Recurrent Neural Network (RNN) & LSTM

207. A 208. D 209. B 210. B 211. A 212. B
213. B 214. B 215. C 216. B 217. C 218. A
219. B 220. B 221. B 222. A 223. B 224. A
225. C 226. B 227. B 228. B 229. D 230. B
231. C 232. C 233. C

Word2vec

234. B 235. C 236. B 237. B 238. B 239. B
240. A 241. B 242. B 243. B 244. A 245. B
246. B 247. False 248. B 249. B 250. B 251. A
252. A 253. A 254. B 255. A 256. B 257. B
258. A 259. B 260. C 261. B 262. B 263. B
264. B 265. B 266. B 267. A 268. B 269. B
270. A 271. A

Answer Key — Part D (Quiz 5 & Quiz 6)

Quiz 5

272. Answer: C — 463

Simple explanation:

The code computes `size = int(len(brown_tagged_sents) * 0.9)`, and we are told this equals **4160**. Since `train_sents = brown_tagged_sents[:size]` and `test_sents = brown_tagged_sents[size:]`, the training set is the first 4160 sentences (90%) and the testing set is everything else (the remaining ~10%).

Step 1 — Find the total number of sentences:

Since $\text{size} = 0.9 \times \text{total} \approx 4160$, we get $\text{total} \approx 4160 / 0.9 \approx 4622.2$.

Because Python's `int()` always rounds *down* (truncates), several values of "total" close to 4622 could all truncate down to exactly 4160 once multiplied by 0.9. We need the value of "total" that makes this exact.

Step 2 — Find the exact total by testing nearby values:

- If total = 4622: $\text{int}(4622 \times 0.9) = \text{int}(4159.8) = 4159$ ✗ (not 4160)
- If total = 4623: $\text{int}(4623 \times 0.9) = \text{int}(4160.7) = 4160$ ✓ (matches!)

Step 3 — Calculate the testing set size:

testing size = total – training size = 4623 – 4160 = **463**

This is why the answer is 463 rather than the "expected" 462 you'd get from a simple $4160 / 0.9$ calculation — the rounding-down behavior of `int()` shifts the true total up by one sentence.

273. Answer: B — 90% for testing and 10% for training

Simple explanation:

This question uses the same numbers as Question 272, but the slicing is *swapped*:

- `train_sents = brown_tagged_sents[size:]` (everything *after* position 4160)
- `test_sents = brown_tagged_sents[:size]` (everything *before* position 4160, i.e. the first 4160 sentences)

Since **size = 4160 already represents 90% of the data** (as established in Question 272), and now it is the *test_sents* that takes the first "size" sentences, it means:

- `test_sents` = 90% of the data
- `train_sents` = the remaining 10% of the data

So the split is **90% for testing and 10% for training** — the reverse of what you'd normally expect. The key trick in this question is paying close attention to which variable (`train_sents` or `test_sents`) is assigned the larger "size" slice, not just recalculating the percentages.

Quiz 6

274. (i) → LSTM (ii) → Feed Forward Network (iii) → RNN

275. False

276. True